ShelfSense AI - Phase 3 MVP Report

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1. Executive Summary

ShelfSense is a grocery forecasting and inventory planning MVP designed to help store managers make short-term ordering decisions. Grocery demand can change a lot around holidays, promotions, payday periods, weather changes, and local events. Because of this, managers who only use simple averages or intuition may order too much on slow days or too little before high-demand periods.

For Phase 3, we expanded the Phase 2 forecasting prototype into a working planning tool. The user can select a store and product family, choose either a 7-day or 14-day forecast, enter current stock information, and receive a demand forecast with a recommended order quantity.

The current MVP does four main things:

- predicts store-level customer traffic
- predicts unit sales for each store and product family
- shows possible demand drivers such as holidays, events, promotions, and heat
- recommends order quantities rounded to a case-pack size

The main goal of this phase was to show the full workflow from data generation, to model training, to a dashboard that a manager could actually use for planning.

2. User and Use Case

The target user is a grocery store manager or inventory planner who needs to make short-horizon ordering decisions for categories such as beverages, produce, snacks, dairy, bakery, or frozen goods.

A typical workflow would be:

1. The manager selects `S002 / beverages`.
2. The manager chooses a 7-day planning horizon.
3. The manager enters the current stock on hand and case-pack size.
4. ShelfSense returns projected demand, expected traffic, likely demand drivers, and an order recommendation.

5. The manager exports the daily order plan for review.

The main MVP feature is:

A grocery manager can select a store and product family, and ShelfSense returns a demand forecast, a traffic alert, and a recommended inventory order.

3. System Design

The system is split into three main parts.

The first part is the data layer. This includes generated historical sales, transaction counts, store profiles, calendar features, promotions, local events, and weather-related features.

The second part is the model layer. We trained XGBoost regression models for traffic and sales forecasting. These models were compared against simpler baselines so we could check whether the added event and calendar features were actually useful.

The third part is the product layer. This is the Gradio dashboard, which turns the model forecasts into a format that is easier for a store manager to understand and use.

4. Data

The MVP uses a calibrated synthetic Arizona grocery dataset. We used synthetic data so the project would be fully reproducible and would not depend on private retailer data. The dataset was designed to match the original ShelfSense idea more closely than a completely random dataset.

The dataset includes:

- 4 store profiles: Tempe, Phoenix, Mesa, and Scottsdale
- 6 product families: produce, beverages, snacks, dairy, bakery, and frozen
- 731 historical days from 2024-01-01 to 2025-12-31
- 17,544 sales rows
- 2,924 transaction rows
- 14 future days for the forecast horizon

The features include:

- store and product family
- daily unit sales
- store-level transactions
- holiday flag and holiday name
- local event flag and event name
- promotion flag

- weekend and payday flags
- average temperature and heatwave flag
- lag and rolling demand features

The data pipeline is located in `src/generate_synthetic_data.py`. The data notes are documented in `data/data_notes.md`.

The biggest limitation of the data is that it is still synthetic. It is useful for building and testing the MVP, but it does not prove that the model would perform the same way on real grocery sales data. A production version would need to replace this with actual store sales, inventory records, and live holiday/event data.

5. Models

ShelfSense uses two supervised regression models.

The first model predicts daily store-level transactions. This gives the dashboard a traffic forecast, which helps explain whether demand is expected to be higher or lower than normal.

The second model predicts daily unit sales for each store and product family. This forecast is then used by the inventory recommendation engine.

The models use:

- XGBoost regressors
- numeric and categorical preprocessing
- lag features and rolling averages
- calendar, holiday, event, promotion, payday, and weather features

We compared three model versions:

- seasonal naive 7-day lag baseline
- XGBoost without event features
- XGBoost with event, promotion, holiday, payday, and weather features

The train/test split was based on time. The final 56 days were held out as the test period. We did not use a random row split because the real use case is forecasting future dates from past data.

6. Evaluation

The event-aware XGBoost model performed best for both sales and transaction forecasting.

Target	Best model	WMAPE	MAE	RMSE
Sales	Event-aware XGBoost	0.0674	11.82	15.97
Transaction	Event-aware XGBoost	0.0382	38.32	50.62

This comparison was important because one of the main ideas behind ShelfSense is that demand changes around events, holidays, promotions, payday periods, and weather. The XGBoost model without event features still performed well, but the event-aware version gave better results and made the forecasts easier to explain.

From the forecast plots, the model followed the short-term demand trends better than the seasonal baseline. The dashboard also showed useful explanations for demand changes, such as holidays, local events, promotions, and heatwave days.

The results should still be viewed as prototype results because the dataset is synthetic. However, the evaluation shows that the full forecasting workflow is functioning and that the event-related features are helping the model in this controlled scenario.

7. MVP Demo

The runnable MVP is `app.py`.

The app includes:

- store selector
- product-family selector
- 7-day or 14-day horizon selector
- current-stock input
- service-level selector
- case-pack input
- KPI cards
- demand and traffic forecast plots
- daily inventory plan
- downloadable CSV export

The demo can be run locally with:

```
cd C:\Users\gaelb\Downloads\shelfsenephase2\mvp
powershell -ExecutionPolicy Bypass -File .\run_all.ps1
```

C:\Users\gaelb\AppData\Local\Programs\Python\Python312\python.exe app.py

The main purpose of the demo is to show that the project is more than just a model. The user can enter planning information, view the forecast, and download an order plan.

8. Limitations and Risks

The MVP works as a course prototype, but there are still several important limitations.

The largest limitation is the use of synthetic data. The dataset was built to behave like an Arizona grocery scenario, but it has not been validated against a real retailer's sales history. Because of that, the accuracy values should not be interpreted as proof of real-world performance.

The event data is also simplified. Local events are currently generated as proxy events instead of being pulled from a live event API. This helps test the workflow, but it is not the same as using real event schedules.

The inventory logic is also simplified. Current stock is entered manually, and supplier lead times are not modeled in detail. A real system would need to account for delivery schedules, minimum order quantities, shelf capacity, spoilage, and supplier constraints.

Another limitation is that the forecasts are at the product-family level instead of the SKU level. This makes the MVP easier to build and understand, but real grocery ordering usually happens at a more detailed item level.

Finally, the explanation features should not be treated as causal proof. The dashboard can point out likely drivers such as holidays, promotions, heat, or events, but it does not prove that one specific factor caused a demand increase.

Privacy risk is low for this prototype because no customer-level data is used.

9. Next Steps

With 2-3 more months, the first priority would be replacing the synthetic data with a real public grocery dataset such as Favorita or M5. This would make the model evaluation more meaningful and would help test whether the same approach works outside of the generated scenario.

The next priority would be improving the event and calendar inputs. A future version could use APIs such as Nager.Date, Ticketmaster, or similar sources to pull real holidays and local events.

Other possible improvements include:

- adding SKU-level forecasting

- adding perishable-item constraints
- adding prediction intervals or confidence bands
- connecting recommendations to stockout and waste costs
- improving lead time and supplier order logic
- deploying the app through Hugging Face Spaces
- running small user tests with students or retail workers acting as store managers

These improvements would make ShelfSense closer to a real inventory planning system instead of just a course MVP.

10. Conclusion

Phase 3 produced a working MVP for ShelfSense. The project now includes a synthetic data pipeline, trained forecasting models, benchmark results, a Gradio dashboard, and inventory order recommendations.

The main weakness is that the model is still based on synthetic data, so the results should be viewed as prototype results rather than proof that the system would work the same way for a real grocery store. Even with that limitation, the MVP shows the intended workflow from forecasting demand to recommending an order quantity.